PROMPT ENGINEERING TUTORIAL

Logic · Technology · Bad Prompts · Good Prompts · Outputs

COMPLETE REFERENCE GUIDE · 23 CORE TECHNIQUES

23	6	100+
Techniques	Core Concepts	Examples

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CORE TECHNOLOGIES

Modern language models are built on **transformer architectures** that predict the next token based on previous tokens, context, attention patterns, and probability distributions. Prompt engineering works by reshaping those probability distributions.

Technology	Role in Prompt Engineering
Tokens	AI processes sub-word units; wording changes tokenization and behavior
Transformers	Attention mechanisms determine which previous words matter most
Probability	AI predicts the next token; prompts reshape the probability space
Context Window	Limited working memory — large irrelevant prompts reduce performance

The Prompt Control Model

Layer	What It Controls
Context	What the AI sees
Goal	What the AI optimises
Constraints	What the AI avoids
Style	How the AI sounds
Structure	How the AI formats output
Reasoning	How the AI thinks

1 SPECIFICITY PROMPTING

Logic: *Specific prompts reduce ambiguity by narrowing token probability distributions.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Explain gravity.`	`Explain gravity to a 10-year-old using football and jumping examples in under 150 words.`

OUTPUTS

`Gravity is a force that pulls objects together. [Too generic]`	`Gravity is the invisible force that pulls everything toward Earth. When you kick a football upward, gravity pulls it back down. When you jump, gravity brings you back to the ground instead of letting you float away like astronauts in space.`

2 ROLE PROMPTING

Logic: *Roles activate profession-specific patterns. The model learned statistical patterns from billions of role-based texts.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Write about climate change.`	`You are a NASA scientist briefing government officials about climate change risks.`

OUTPUTS

`Climate change is a major issue affecting the planet. [Generic and shallow]`	`Rising atmospheric CO2 levels are increasing global temperatures, intensifying storms, droughts, and coastal flooding. Immediate investment in mitigation and adaptation infrastructure is recommended.`

3 OUTPUT FORMATTING

Logic: *If format is unspecified, AI guesses. Transformers imitate structural patterns from training data.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Compare Python and JavaScript.`	`Compare Python and JavaScript in a table with columns: syntax, speed, ecosystem, best use cases.`

OUTPUTS

`Python and JavaScript are both programming languages. Python is simpler while JavaScript is used in websites.`	`Feature | Python | JavaScript Syntax | Simple | Flexible Speed | Moderate | Fast in browser Ecosystem | Strong AI/Data | Strong web tools Best Use | AI, automation | Frontend, backend`

4 CHAIN-OF-THOUGHT PROMPTING

Logic: *Intermediate reasoning improves complex problem solving — reasoning tokens stabilise computation pathways.*

✗ BAD PROMPT	✓ GOOD PROMPT
`A train travels 60 km in 1 hour. How far in 5 hours?`	`Solve step by step: A train travels 60 km in 1 hour. How far in 5 hours?`

OUTPUTS

`240 km [Incorrect — skipped reasoning]`	`Distance in 1 hour = 60 km In 5 hours: 60 × 5 = 300 km Final answer: 300 km`

5 ZERO-SHOT PROMPTING

Logic: *No examples are given. The model relies entirely on pretrained knowledge to infer the task.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Make this professional: "hey send me the file"`	`Rewrite professionally: "hey send me the file"`

OUTPUTS

`Send me the file. [Minimal improvement]`	`Could you please send me the file at your earliest convenience?`

6 ONE-SHOT PROMPTING

Logic: *A single example defines the expected pattern for output format and style.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Translate: Dog`	`Example: Cat → Gato Now translate: Dog →`

OUTPUTS

`Perro [Correct but format inconsistent]`	`Dog → Perro`

7 MULTI-SHOT PROMPTING

Logic: *Multiple examples establish patterns. Attention identifies repeated mappings across examples.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Classify: Rose`	`Examples: Cat → Animal Rose → Flower Carrot → Vegetable Now classify: Apple →`

OUTPUTS

`Plant [Too broad]`	`Apple → Fruit`

8 DELIMITER PROMPTING

Logic: *Separating tasks reduces confusion. Transformers strongly detect structural boundaries.*

✗ BAD PROMPT	✓ GOOD PROMPT
`summarize translate analyze sentiment classify keywords`	`ARTICLE: """ AI is transforming education. """ Tasks: 1. Summarize 2. Translate to Hindi 3. Analyse sentiment`

OUTPUTS

`Messy mixed response with no clear boundaries.`	`Summary: AI is changing education. Hindi: AI shiksha ko badal raha hai. Sentiment: Positive`

9 CONSTRAINT PROMPTING

Logic: *Constraints reduce drift by suppressing irrelevant token paths.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Write an email requesting leave.`	`Write a professional leave request email under 80 words.`

OUTPUTS

`Long unfocused email with unnecessary padding.`	`Dear Manager, I would like to request leave for May 28 due to personal reasons. Please let me know if any additional information is needed. Regards, Rahul`

10 AUDIENCE PROMPTING

Logic: *Different audiences need different abstraction levels. Matching vocabulary to the reader is essential.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Explain quantum mechanics.`	`Explain quantum mechanics to a 12-year-old using toy and game analogies.`

OUTPUTS

`Quantum mechanics studies wave functions and probabilistic operators. [Too technical]`	`Imagine a spinning coin hidden under a cup. Until you look, it's not fully heads or tails. Quantum particles behave somewhat like that.`

11 NEGATIVE PROMPTING

Logic: *Specifying exclusions improves control by restricting the output space.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Explain AI.`	`Explain AI without jargon, equations, or technical terms.`

OUTPUTS

`Very technical explanation filled with ML terminology.`	`AI is like teaching computers to learn patterns and make decisions similarly to how humans learn from experience.`

12 CONTEXT INJECTION

Logic: *AI only knows what is in its context window. Providing relevant data prevents hallucination.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Improve this code.`	`Improve readability only: for i in range(10):print(i)`

OUTPUTS

`Please provide the code. [Failed — no context given]`	`for i in range(10): print(i)`

13 MULTI-STEP DECOMPOSITION

Logic: *Breaking tasks into ordered stages improves reliability and traceability.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Build a startup strategy.`	`Step 1: Identify target audience Step 2: Define pricing Step 3: Create marketing strategy`

OUTPUTS

`Vague generic business advice.`	`Audience: Small businesses Pricing: Subscription model Marketing: LinkedIn ads and webinars`

14 RETRIEVAL-AUGMENTED (RAG)

Logic: *Inserting external documents into context reduces hallucinations and grounds the model in facts.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Explain our HR policy.`	`Using this HR policy document, summarise maternity leave rules.`

OUTPUTS

`Generic HR explanation not relevant to the company.`	`Employees are eligible for 26 weeks of paid maternity leave after completing one year of employment.`

15 STYLE TRANSFER PROMPTING

Logic: *AI can imitate statistical writing styles learned from millions of genre-specific texts.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Rewrite dramatically.`	`Rewrite in the style of a suspense thriller using short tense sentences.`

OUTPUTS

`Unclear inconsistent style with no discernible tone.`	`The lights flickered. Footsteps echoed behind him. He stopped breathing.`

16 ITERATIVE PROMPTING

Logic: *Outputs improve through progressive refinement with explicit revision criteria.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Make this better.`	`Reduce repetition, make tone professional, and shorten by 20%.`

OUTPUTS

`Minimal unclear edits with no direction.`	`Cleaner, concise, polished version that meets all stated criteria.`

17 PROMPT CHAINING

Logic: *One prompt's output feeds the next. Chaining decomposes complex multi-objective tasks.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Analyse this article and generate interview questions and social posts and summaries.`	`Prompt 1: Extract key topics. Prompt 2: Generate interview questions from those topics.`

OUTPUTS

`Mixed chaotic output attempting all goals at once.`	`P1 Output: AI ethics, automation, education` `P2 Output: How should AI ethics be regulated?`

18 REFLECTION PROMPTING

Logic: *AI critiques its own output and iterates, catching logical gaps and clarity issues.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Write an essay on renewable energy.`	`Write the essay, then review it for logical gaps and improve clarity.`

OUTPUTS

`Essay with repetition, weak structure, and unsupported claims.`	`More coherent and refined essay with improved logical flow and stronger evidence.`

19 SELF-CONSISTENCY PROMPTING

Logic: *Generating multiple reasoning paths and comparing them improves robustness of the answer.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Solve this equation once.`	`Solve using algebra and graphing, then compare answers.`

OUTPUTS

`Potential reasoning mistake from a single solution path.`	`Both methods produce the same result — increases confidence in correctness.`

20 TOOL-USE PROMPTING

Logic: *External tools (search, calculator, API) improve factual accuracy beyond the training cutoff.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Guess today's stock price.`	`Use live market data to analyse today's stock movement.`

OUTPUTS

`Hallucinated value with no grounding.`	`Accurate current market analysis based on real-time data.`

21 GOAL-ORIENTED PROMPTING

Logic: *Framing around the desired outcome produces more targeted and actionable results.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Write a LinkedIn post.`	`Write a LinkedIn post designed to attract AI recruiters for machine learning jobs.`

OUTPUTS

`Generic motivational post with no targeting.`	`Targeted professional networking content with relevant ML keywords and calls to action.`

22 TEMPERATURE-AWARE PROMPTING

Logic: *Different tasks require different creativity levels. Legal and factual tasks need conservative language.*

✗ BAD PROMPT	✓ GOOD PROMPT
`Generate a legal contract creatively.`	`Generate a precise formal legal summary using conservative language.`

OUTPUTS

`Unstable inconsistent wording unsuitable for legal use.`	`Stable professional legal-style text with precise, unambiguous phrasing.`

23 CONSTRAINT HIERARCHY

A strong prompt follows a consistent priority order:

Priority	Component	Purpose
1 — Highest	Objective	Define what success looks like
2	Constraints	Specify what to avoid or limit
3	Context	Provide relevant background information
4	Style	Set tone, voice, and register
5 — Lowest	Formatting	Specify output structure

! COMMON PROMPT ENGINEERING MISTAKES

Mistake	Problem	Fix
Vagueness	Random, unpredictable outputs	Add specificity and constraints
Too many tasks	Confusion and mixed responses	Use prompt chaining

Missing examples	Inconsistent format	Add one-shot or few-shot examples
No constraints	Output drift	Define length, tone, exclusions
Wrong audience	Wrong complexity level	Specify audience explicitly
Giant prompts	Attention dilution	Remove irrelevant context

∞ UNIVERSAL PROMPT TEMPLATE

ROLE	`You are an expert [profession/persona]...`
TASK	`Perform this task: [clear specific objective]...`
CONTEXT	`Relevant background information: [domain knowledge, data]...`
CONSTRAINTS	`Length: [X words]. Exclude: [topics]. Style: [tone/voice]...`
OUTPUT FORMAT	`Table / JSON / Bullet points / Paragraph...`
QUALITY CHECK	`Review for errors before finalising the response.`

→ THE FUTURE OF PROMPT ENGINEERING

Prompt engineering is evolving beyond 'prompt tricks.' Modern AI systems increasingly combine **prompting**, **retrieval**, **memory**, **planning**, **tool usage**, **reflection**, **reasoning**, and **autonomous workflows**. The field is moving toward **Human-AI cognitive system design** — less about clever wording and more about designing structured reasoning systems.

Capability	Description
Prompting	Crafting precise instructions to guide model behaviour
Retrieval (RAG)	Grounding outputs in external, up-to-date documents
Memory	Maintaining state across multi-turn interactions
Planning	Decomposing goals into ordered subtasks
Tool Usage	Calling APIs, calculators, search engines
Reflection	Self-critique and iterative improvement of outputs
Reasoning	Multi-step logical inference and chain-of-thought
Autonomous Workflows	Agents that plan, act, observe, and iterate independently